Trading Horizon as Endogenous Decision

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1 Theory

1.1 Trading Horizon as an Endogenous Decision

Consider an investor trading a collection of assets indexed by $i \in \{1, \ldots, N\}$. At any decision time t, the investor may express a position over one of several admissible trading horizons. Let $\mathcal{H} = \{h_1, h_2, \ldots, h_K\}$ denote a finite, discrete set of horizons. A horizon is not just a sampling convention, it defines the interval over which a trading decision is intended to earn a return and over which its risk and implementation cost are realized.

For each horizon $h \in \mathcal{H}$, let $q_{i,t}^{(h)} \in [-1, 1]$ denote the signed fraction of the capital assigned to horizon h that is exposed to asset i at time t. Positive values represent long positions and negative values represent short positions. Thus the choice is real-valued: the investor may assign partial long or short exposure to an asset rather than making only an all-or-nothing decision.

The investor has finite available capital, normalized to one. Let $a_{h,t} \in [0, 1]$ denote the fraction of total capital allocated to trading decisions at horizon h. Capital allocations must satisfy

$$\sum_{h \in \mathcal{H}} a_{h,t} \leq 1, \qquad \sum_{i=1}^{N} \left| q_{i,t}^{(h)} \right| \leq 1 \quad \text{for each } h \in \mathcal{H}.$$

The absolute-value restriction bounds gross exposure within each horizon allocation, allowing offsetting long and short positions without implicitly permitting unlimited leverage. Any unallocated share may be understood as capital held outside the risky positions. The construction of the candidate exposures is left unrestricted at this stage. The resulting signed fraction of total capital exposed to asset i is

$$Q_{i,t} = \sum_{h \in \mathcal{H}} a_{h,t} q_{i,t}^{(h)}.$$

Let $r_{i,t,t+h}$ be the return of asset i over the interval associated with horizon h, and let $C_{t,t+h}^{(h)}$ represent the costs incurred by implementing the corresponding horizon-specific allocation. The net payoff attributable to horizon h is

$$\Pi^{(h)}_{t,t+h} = a_{h,t} \sum_{i=1}^{N} q^{(h)}_{i,t} r_{i,t,t+h} - C^{(h)}_{t,t+h}.$$

Therefore the investor's decision is not only which assets to trade, but also which trading horizons should receive capital. Distinct horizons may offer different expected payoffs, risks, costs, and dependence with one another. An allocation that ignores the horizon dimension implicitly assumes that the same trading window is always appropriate, or that the relative value of alternative windows is constant.

The basic hypothesis of this framework is that trading horizon is an endogenous portfolio choice. Conditional on the information available at time t, the allocation vector $a_t = (a_{h,t})_{h\in\mathcal{H}}$ is chosen jointly with the real-valued asset exposure choices, subject to the investor's finite capital and risk constraints. The objective is to allocate exposure to the set of horizon-specific opportunities that provides the most attractive net risk-adjusted payoff.

1.2 Decision Times and Overlapping Horizon Positions

Let $\delta > 0$ denote the shortest interval at which the investor may revise the portfolio. Decision times lie on the common grid $\mathcal{T} = \{t_0 + n\delta : n = 0, 1, \ldots, M\}$. The admissible horizons are aligned with this grid: $\mathcal{H} \subseteq \{m\delta : m \in \mathbb{N}\}$. Consequently, for every $t \in \mathcal{T}$ and every $h \in \mathcal{H}$, both the initiation time t and the maturity time $t + h$ lie on the investor's decision grid.

At each decision time t, the investor may initiate a new position for every admissible horizon, including a new longer-horizon position even when earlier positions at that same horizon remain active. Define the newly initiated signed exposure to asset i for the position vintage (t, h) as $x^{(h)}_{i,t} = a_{h,t} q^{(h)}_{i,t}$. A vintage initiated at time t for horizon h remains active until $t + h$. Thus, positions initiated at different decision times can have overlapping payoff intervals and distinct maturity times. At any decision time $s \in \mathcal{T}$, total active exposure to asset i is

$$Q_{i,s} = \sum_{h\in\mathcal{H}} \sum_{\substack{t\in\mathcal{T} \\ t\le s<t+h}} x^{(h)}_{i,t}.$$

Two notions of capacity are relevant when horizon-specific positions overlap. First, let $G^{\text{exec}} > 0$ denote the admissible gross exposure of the position actually executed in the market after active vintages are netted within each asset. Feasible executed exposure satisfies

$$\sum_{i=1}^{N} |Q_{i,s}| \le G^{\text{exec}}, \qquad s \in \mathcal{T}.$$

Second, let $B^{\text{commit}} > 0$ denote the permitted aggregate commitment to all active horizon-specific decisions before netting. This is not a broker-level capital constraint; it is an overlap-control restriction that limits simultaneous, potentially conflicting, horizon-specific exposure. Active decisions must satisfy

$$\sum_{i=1}^{N} \sum_{h \in \mathcal{H}} \sum_{\substack{t \in \mathcal{T} \\ t \leq s < t+h}} \left| x_{i,t}^{(h)} \right| \leq B^{\text{commit}}, \qquad s \in \mathcal{T}.$$

With normalized capital and no leverage, G^{exec} may be set to one; B^{commit} is chosen as a separate modeling limit. The two restrictions serve different purposes: the first limits the broker-level portfolio after offsetting positions are combined, while the second prevents opposing position vintages from being added without limit merely because they net to a small executed exposure. Thus overlapping horizon decisions do not provide new capital. A position associated with a longer horizon continues to consume active decision commitment capacity when newer decisions are made.

1.3 The Dynamic Allocation Problem

Let W_t denote investor wealth immediately before the decision at time t, and let $x_t = \left(x_{i,t}^{(h)} \right)_{\substack{i=1,\ldots,N \\ h \in \mathcal{H}}}$ collect the new horizon-specific exposures selected at that time. The investor chooses x_t conditional on the information available at t and on all position vintages that remain active from earlier decisions. In particular, x_t may include a newly initiated longer-horizon position even if positions with the same horizon were initiated at earlier decision times and have not yet matured. In general form, the decision problem is

$$x_t^* \in \arg\max_{x_t \in \mathcal{A}_t} \left\{ \mathbb{E}_t[\Delta W(x_t)] - \gamma \mathcal{R}_t(x_t) - \mathcal{C}_t(x_t; Q_t^{\text{existing}}) \right\},$$

where $\mathbb{E}_t[\Delta W(x_t)]$ is the conditional expected payoff from the decision, $\mathcal{R}_t(x_t)$ is its risk contribution, $\mathcal{C}_t(x_t; Q_t^{\text{existing}})$ is the implementation and adjustment cost given the existing portfolio, and $\gamma \geq 0$ controls risk aversion.

The feasible set enforces both capacity constraints after combining the new decision with active exposures. Let $\mathcal{V}_t^{\text{existing}}$ be the set of position vintages initiated prior to t that have not matured. Then

$$\mathcal{A}_t = \left\{ x_t : \sum_{i=1}^{N} \left| Q_{i,t}^{\text{existing}} + \sum_{h \in \mathcal{H}} x_{i,t}^{(h)} \right| \leq G^{\text{exec}}, \quad \sum_{i=1}^{N} \left(\sum_{(\tau,h) \in \mathcal{V}_t^{\text{existing}}} \left| x_{i,\tau}^{(h)} \right| + \sum_{h \in \mathcal{H}} \left| x_{i,t}^{(h)} \right| \right) \leq B^{\text{commit}} \right\}.$$

The investor may consequently revise the allocation across trading horizons at every decision time, but only within the portfolio's executed-exposure and active-decision commitment limits and after accounting for active positions, risk, and adjustment cost.